Voltage & Overcurrent Protector Using Arduino UNO

FSMB Recruitment 2026 - Phase 1 | Embedded System Engineer | Simulation Report

Objective and approach

I designed a simulated Arduino UNO based protector to disconnect a low voltage stand in load when either the voltage or current exceeds a adjustable limit. The design also supports serial communication with a computer and a manual reset after faults.

Circuit and implementation

Two potentiometers supply analog signals to A0 (voltage) and A1 (current). Their 0–1023 ADC readings are mapped to simulated ranges of 0–300 V and 0–15 A. A relay module on D7 switches a 5 V yellow LED load through COM/NO contacts. D8 and D9 drive normal/fault LEDs and a push button on D2 provides manual reset.

Arduino firmware samples the analog inputs approximately every 50 ms and trips the relay if voltage exceeds 250 V or current exceeds 10 A (initial settings). A latched fault keeps the load disconnected until the readings return to safe levels and a reset is requested.

Computer interface

The USB serial interface operates at 9600 baud. STATUS displays measurements and the relay/fault state; SETV and SETI modify the protection limits; RESET requests a safe reconnection. HELP lists available commands and pauses periodic status reports; a valid threshold update or reset command resumes the reports.

Simulation tests and observations

I exercised normal conditions, overvoltage, overcurrent and manual reset in Wokwi. The simulated load LED remained on during safe readings and turned off when protection tripped; normal operation was restored after safe readings and reset. Test screenshots are provided in the test_evidence folder.

The project_files/diagram.json file contains the Wokwi circuit configuration and src/sketch.ino contains the firmware.

Limitations and next steps

All voltage and current readings are emulated by potentiometers: no real grid sensor, real appliance or mains connection was used. This simulation does not validate physical switching speed, surge endurance, electrical isolation or production safety. Settings are volatile and return to defaults on restart. Future hardware work would need isolated sensing, a rated switching stage and controlled electrical safety tests.

Simulation-only assessment | Wokwi + Arduino UNO | No mains-voltage connection